Practical Model Examination

Name: Joshika Saravanan Course-code: CSA0504

Reg.no: 192471009

Course: Database Management System

SET -10

Hotel Management System

Scenario

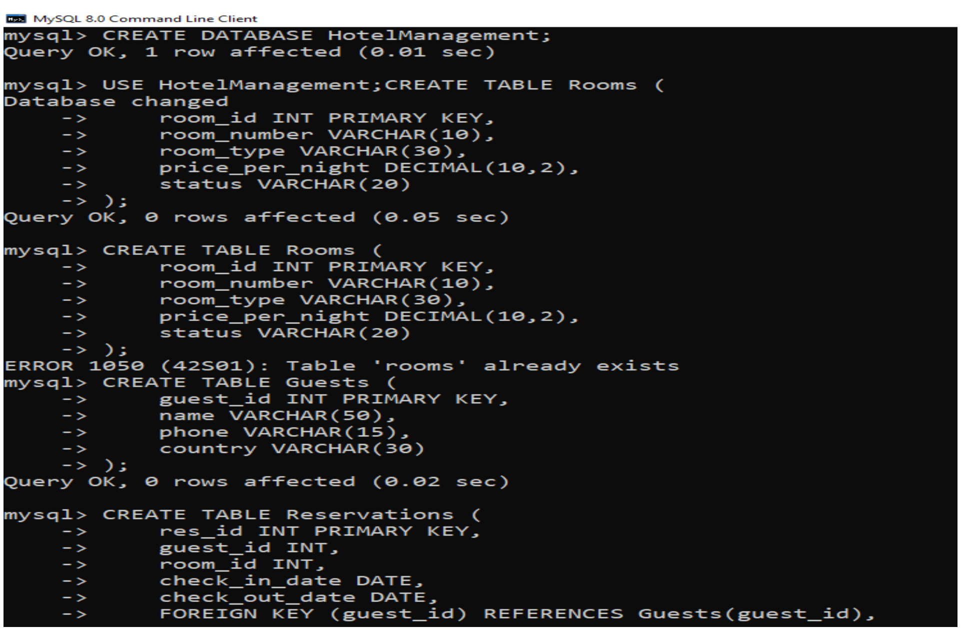
A hotel manages room allocations, reservations, and customer billing.

•Rooms (room_id, room_number, room_type, price_per_night, status)

•Guests (guest_id, name, phone, country)

•Reservations (res_id, guest_id, room_id, check_in_date, check_out_date)

•Payments (payment_id, res_id, amount_paid, payment_method, payment_date)



```
MySQL 8.0 Command Line Client
mysql> CREATE DATABASE HotelManagement;
Query OK, 1 row affected (0.01 sec)

mysql> USE HotelManagement;CREATE TABLE Rooms (
Database changed
    ->     room_id INT PRIMARY KEY,
    ->     room_number VARCHAR(10),
    ->     room_type VARCHAR(30),
    ->     price_per_night DECIMAL(10,2),
    ->     status VARCHAR(20)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE Rooms (
    ->     room_id INT PRIMARY KEY,
    ->     room_number VARCHAR(10),
    ->     room_type VARCHAR(30),
    ->     price_per_night DECIMAL(10,2),
    ->     status VARCHAR(20)
    -> );
ERROR 1050 (42S01): Table 'rooms' already exists
mysql> CREATE TABLE Guests (
    ->     guest_id INT PRIMARY KEY,
    ->     name VARCHAR(50),
    ->     phone VARCHAR(15),
    ->     country VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE Reservations (
    ->     res_id INT PRIMARY KEY,
    ->     guest_id INT,
    ->     room_id INT,
    ->     check_in_date DATE,
    ->     check_out_date DATE,
    ->     FOREIGN KEY (guest_id) REFERENCES Guests(guest_id),
```

MySQL 8.0 Command Line Client

```
    ->     FOREIGN KEY (room_id) REFERENCES Rooms(room_id)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE Payments (
    ->     payment_id INT PRIMARY KEY,
    ->     res_id INT,
    ->     amount_paid DECIMAL(10,2),
    ->     payment_method VARCHAR(30),
    ->     payment_date DATE,
    ->     FOREIGN KEY (res_id) REFERENCES Reservations(res_id)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Rooms VALUES
    -> (1, '101', 'Single', 2000, 'Available'),
    -> (2, '102', 'Double', 3000, 'Available'),
    -> (3, '103', 'Deluxe', 5000, 'Available'),
    -> (4, '104', 'Suite', 8000, 'Available'),
    -> (5, '105', 'Single', 2000, 'Available');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Guests VALUES
    -> (1, 'Arun', '9876543210', 'India'),
    -> (2, 'Priya', '9876543211', 'India'),
    -> (3, 'John', '9876543212', 'USA');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Reservations VALUES
    -> (1, 1, 1, '2026-09-01', '2026-09-05'),
    -> (2, 2, 2, '2026-09-02', '2026-09-06'),
    -> (3, 3, 3, '2026-08-20', '2026-08-25');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

Select MySQL 8.0 Command Line Client

```
    ->     payment_date DATE,
    ->     FOREIGN KEY (res_id) REFERENCES Reservations(res_id)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Rooms VALUES
    -> (1, '101', 'Single', 2000, 'Available'),
    -> (2, '102', 'Double', 3000, 'Available'),
    -> (3, '103', 'Deluxe', 5000, 'Available'),
    -> (4, '104', 'Suite', 8000, 'Available'),
    -> (5, '105', 'Single', 2000, 'Available');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Guests VALUES
    -> (1, 'Arun', '9876543210', 'India'),
    -> (2, 'Priya', '9876543211', 'India'),
    -> (3, 'John', '9876543212', 'USA');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Reservations VALUES
    -> (1, 1, 1, '2026-09-01', '2026-09-05'),
    -> (2, 2, 2, '2026-09-02', '2026-09-06'),
    -> (3, 3, 3, '2026-08-20', '2026-08-25');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Payments VALUES
    -> (1, 1, 8000, 'UPI', '2026-09-01'),
    -> (2, 2, 12000, 'Card', '2026-09-02'),
    -> (3, 3, 25000, 'Cash', '2026-08-20');
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> // question 1
```

Questions

1. **SQL & Joins**: Write a query using RIGHT JOIN or FULL OUTER JOIN to display all room details along with reservation details, ensuring rooms that have never been reserved are also listed.

```
SELECT
    r.room_id,
    r.room_number,
    r.room_type,
    r.p' at line 1
mysql> SELECT
    ->     r.room_id,
    ->     r.room_number,
    ->     r.room_type,
    ->     r.price_per_night,
    ->     r.status,
    ->     rs.res_id,
    ->     rs.guest_id,
    ->     rs.check_in_date,
    ->     rs.check_out_date
    -> FROM Reservations rs
    -> RIGHT JOIN Rooms r
    -> ON rs.room_id = r.room_id;
+---------+-------------+-----------+-----------------+-----------+--------+----------+---------------+----------------+
| room_id | room_number | room_type | price_per_night | status    | res_id | guest_id | check_in_date | check_out_date |
+---------+-------------+-----------+-----------------+-----------+--------+----------+---------------+----------------+
|       1 | 101         | Single    |         2000.00 | Available |      1 |        1 | 2026-09-01    | 2026-09-05     |
|       2 | 102         | Double    |         3000.00 | Available |      2 |        2 | 2026-09-02    | 2026-09-06     |
|       3 | 103         | Deluxe    |         5000.00 | Available |      3 |        3 | 2026-08-20    | 2026-08-25     |
|       4 | 104         | Suite     |         8000.00 | Available |   NULL |     NULL | NULL          | NULL           |
|       5 | 105         | Single    |         2000.00 | Available |   NULL |     NULL | NULL          | NULL           |
+---------+-------------+-----------+-----------------+-----------+--------+----------+---------------+----------------+
5 rows in set (0.02 sec)
```

2. **Views:** Create a view vw_current_guests displaying guest_name, room_number, room_type, and check_in_date for all active reservations where the current date falls between check_in_date and check_out_date.

```
mysql> CREATE VIEW vw_current_guests AS
    -> SELECT
    ->     g.name AS guest_name,
    ->     r.room_number,
    ->     r.room_type,
    ->     rs.check_in_date
    -> FROM Guests g
    -> JOIN Reservations rs
    -> ON g.guest_id = rs.guest_id
    -> JOIN Rooms r
    -> ON rs.room_id = r.room_id
    -> WHERE CURDATE() BETWEEN rs.check_in_date AND rs.check_out_date;
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM vw_current_guests;
+------------+-------------+-----------+---------------+
| guest_name | room_number | room_type | check_in_date |
+------------+-------------+-----------+---------------+
| Arun       | 101         | Single    | 2026-09-01    |
| Priya      | 102         | Double    | 2026-09-02    |
+------------+-------------+-----------+---------------+
2 rows in set (0.01 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION fn_total_stay_cost(
    ->     p_check_in_date DATE,
    ->     p_check_out_date DATE,
```

3. **User-Defined Function:** Write a function fn_total_stay_cost that accepts check_in_date, check_out_date, and price_per_night, calculates the number of nights, and returns the total stay cost.

Select MySQL 8.0 Command Line Client

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION fn_total_stay_cost(
    ->     p_check_in_date DATE,
    ->     p_check_out_date DATE,
    ->     p_price_per_night DECIMAL(10,2)
    -> )
    -> RETURNS DECIMAL(10,2)
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE total_nights INT;
    ->     DECLARE total_cost DECIMAL(10,2);
    ->
    ->     SET total_nights = DATEDIFF(
    ->         p_check_out_date,
    ->         p_check_in_date
    ->     );
    ->
    ->     SET total_cost = total_nights * p_price_per_night;
    ->
    ->     RETURN total_cost;
    -> END //
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT fn_total_stay_cost(
    ->     '2026-09-01',
    ->     '2026-09-05',
    ->     2000
    -> ) AS total_stay_cost;
+-----------------+
| total_stay_cost |
+-----------------+
|         8000.00 |
+-----------------+
1 row in set (0.01 sec)
```

4. **Triggers:** Write a trigger trg_update_room_status that automatically updates the room status in the Rooms table to 'Occupied' when a new row is inserted into Reservations.

Select MySQL 8.0 Command Line Client

```
mysql> CREATE TRIGGER trg_update_room_status
    -> AFTER INSERT ON Reservations
    -> FOR EACH ROW
    -> BEGIN
    ->     UPDATE Rooms
    ->     SET status = 'Occupied'
    ->     WHERE room_id = NEW.room_id;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT * FROM Rooms;
+---------+-------------+-----------+-----------------+-----------+
| room_id | room_number | room_type | price_per_night | status    |
+---------+-------------+-----------+-----------------+-----------+
|       1 | 101         | Single    |         2000.00 | Available |
|       2 | 102         | Double    |         3000.00 | Available |
|       3 | 103         | Deluxe    |         5000.00 | Available |
|       4 | 104         | Suite     |         8000.00 | Available |
|       5 | 105         | Single    |         2000.00 | Available |
+---------+-------------+-----------+-----------------+-----------+
5 rows in set (0.00 sec)

mysql> INSERT INTO Reservations
    -> VALUES (4, 1, 4, '2026-09-03', '2026-09-07');
Query OK, 1 row affected (0.02 sec)

mysql> SELECT room_id, room_number, status
    -> FROM Rooms
    -> WHERE room_id = 4;
+---------+-------------+----------+
| room_id | room_number | status   |
+---------+-------------+----------+
|       4 | 104         | Occupied |
+---------+-------------+----------+
```